Programming tasks

In Programming task, you will implement linear regression in different approaches as follows:

- Q2: You will implement a linear regression from scratch using Python and PyTorch.
- Q3: You will implement *linear regression* using *sklearn library*
- Q4: You will implement *Normal equation* to get the values of θ

Q2) Given dataset_q2_q4.csv file that contains 500 samples with 3 features (x_1, x_2, x_3) and the expected output (y). The **hypothesis** is defined as follows:

$$h(x)_\theta = 3\theta_1 x_1^2 + \theta_2 x_2^3 - \theta_3 x_2^3 + \theta_2\theta_3 x_3^3 + \theta_0$$

The code to load and split data in X *and* y is given in the provided notebook file. Write Python code to do the following tasks:

a. Split the dataset into 60% train data and 40% test data.

b. Write a program in **two** different ways using **Python** and **PyTorch** to implement linear regression using gradient decent based. The loss function is defined as follows:

$$J(\theta_0, \theta_1, \theta_2, \theta_3) = \frac{1}{2m}\sum_{i=1}^{m}(h_\theta(x^{(i)} - y^{(i)})^2$$

c. Plot the cost/loss function against the iterations (epochs)

d. What are the best values of $\theta_0, \theta_1, \theta_2, and\ \theta_3$ that minimizes the cost function?

Q3) Use *make_regression* function in *skearn* to generate dataset with 800 samples with 15 features with random_state=10. The hypothesis is defined as a linear function

a. Split the dataset into 70% train data and 30% test data.
b. Use **sklearn** to implement the following:
   - Linear regression
   - Lasso regression
   - Ridge regression
   - Polynomial linear regression

c. Draw bar chart or table that compares the mean squared error between the correct outputs and the predicted outputs on testing set for linear models.

Q4) Normal equation (use dataset in **Q3**)

You learned normal equation (closed-form solution) that can be used for finding the optimal parameter values θ for linear regression given the matrix of training examples X and the corresponding response variables y. Write a Python code to implement normal equation. *The formula is defined as follows:*

$$\theta = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$